Fisheries Research 96 (2009) 58–63

Contents lists available at ScienceDirect

Fisheries Research

journal homepage: www.elsevier.com/locate/fishres

Spatial scales of variation of assemblages of fish in coastal lakes sampled with multi-mesh gillnets: Implications for designing research surveys

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ARTICLE INFO

Keywords:
Fishery-independent survey
Pilot study
Standardised sampling
Hierarchical experimental design
Variance estimation
Permanova

ABSTRACT

A key element in designing surveys to assess populations and assemblages of fish is to understand their spatial and temporal scales of variation. As part of a strategy to develop fishery-independent surveys, a pilot experiment was done in two coastal lakes in south-eastern Australia to examine scales of spatial variation among assemblages of fish caught in multi-mesh gillnets. A hierarchical, nested sampling design measured spatial variation among zones (2–20 km apart), sites within zones (1 km apart) and replicate gillnets at each site (50–100 m apart), in shallow and deep habitats. Assemblages differed between lakes. In both lakes, most species were caught in greater numbers in shallow samples, which had greater proportions of small fish of some species. Spatial variation in uni- and multivariate analyses was greatest at the smallest spatial scale (i.e. among replicates), which was consistent across species, depths and lakes. Variation among samples was also generally greater among sites than zones. It would be unnecessary to include the scale of zones in future sampling and more effort should be placed on sampling replicate nets and sites within lakes. The importance of doing pilot experiments and their general applicability to designing surveys of fisheries resources is illustrated.

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1. Introduction

Much research has focused on understanding how fauna and flora vary over a range of spatial and temporal scales (Morrisey et al., 1992; Fraschetti et al., 2005; Stewart-Koster et al., 2007). Measuring spatial and temporal variation is important for understanding the mechanisms and processes that structure assemblages of species, assessing environmental impacts and for identifying appropriate scales of sampling (Levin, 1992; Underwood et al., 2000). Many studies have measured the spatial and temporal variation of organisms using hierarchical analyses of variance (Morrisey et al., 1992; Murdoch and Aronson, 1999), but there are few such studies of fish (Anderson and Millar, 2004). Moreover, few of these studies have tested for consistency of patterns across habitats.

Coastal lakes contain abundant and diverse fish faunas, which support commercial and recreational fisheries (e.g. Albaret and Lae, 2003; Gray and Kennelly, 2003; Rueda and Defeo, 2003). These environments are subject to strong environmental fluctuations (Kjerfve, 1994) and the fish in coastal lakes are also affected by many biotic factors (e.g. recruitment, predation, and competition; Livingston, 1987). Interactions between abiotic and biotic factors in time and space can strongly affect patterns of abundance, causing patchiness at a range of scales (Morrisey et al., 1992).

The problems of sampling at single spatial scales have been well documented (Andrew and Mapstone, 1987). Despite this, fish in lakes and estuarine systems are not generally sampled over multiple spatial scales; sampling is usually stratified across particular sites or habitats (Jones and West, 2005), along environmental gradients, such as salinity or flow regimes (Loneragan et al., 1986), or according to pre-determined spatial grids (Rueda and Defeo, 2003). Thus, potential confounding effects of small-scale variation on larger scale comparisons and, therefore, the appropriateness of many survey designs for sampling fish in lakes and estuaries, remain unknown.

Here, we use a hierarchical sampling design to examine spatial variation in populations and assemblages of fish in two coastal lakes in south-eastern Australia as a key step in developing sampling tools for surveys in coastal lakes and estuaries (Gray et al., 2005; Rotherham et al., 2007). Sampling, including the spatial scales of replicates (50–100 m), sites (1 km) and zones (2–20 km), was done in two depths and in two lakes (separated by hundreds of km) to assess the generality of results. Specifically, we tested the hypotheses that: (i) populations and assemblages vary at each spatial scale examined and (ii) patterns of variation are consistent across depths

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doi:10.1016/j.fishres.2008.09.011

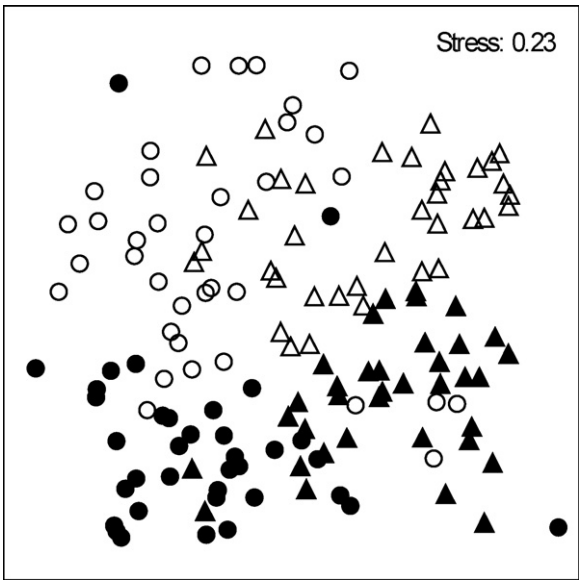

Fig. 1. Two-dimensional nMDS ordination plot of standardised data showing relationships among assemblages of fish sampled with multi-mesh gillnets in Lake Macquarie shallow (○), Lake Macquarie deep (●), St Georges Basin shallow (△) and St Georges Basin deep (▲). Stress value = 0.23.

and lakes. The purpose of the study was to determine what scales should be included in future sampling designs.

2. Methods

2.1. Study locations

Sampling was done in Lake Macquarie (LM) (151°36′E, 33°06′S) and St Georges Basin (SGB) (150°36′E, 35°08′S), in New South Wales (NSW, south-eastern Australia). These lakes are relatively shallow (mean depth <2 m), micro-tidal, well mixed and have no large salinity or tidal gradients. LM has a surface water area of 125 km^2; SGB has a surface water area of 44 km^2 (Roy et al., 2001).

2.2. Sampling design and methods

In each lake, three zones (2–20 km apart) and two sites within each zone (1 km apart), were selected randomly in shallow (<2 m) and deep (4–8 m) habitats. Depths were chosen as potential strata for future surveys. On each night of sampling, 12 replicate multi-mesh gillnets (6 shallow and 6 deep), separated by 50–100 m, were fished at a single site chosen at random. A different site was sampled on each night and the two lakes were sampled over a 13-day period in May/June 2005. Nets were bottom-set at dusk, left to soak for 1 h and then retrieved (Rotherham et al., 2006). Fish caught in each net were identified and counted; economically important species were measured for fork length (FL, to the nearest 0.5 cm).

Each multi-mesh gillnet consisted of 7 m × 20 m panels of different sizes of stretched mesh (36, 44, 54, 63, 76, 89 and 102 mm). The 36 and 44 mm panels were made from monofilament netting with twine diameter 0.15 mm; the remaining panels had multifilament netting with twine diameter 0.4 mm (Gray et al., 2005). All panels were 2 m deep and had a hanging ratio of $E = 0.5$. Adjacent panels were separated by 5 m of rope to maintain independence and restrict the smaller mesh panels 'leading' larger fish to adjacent larger meshed panels. The order of the panels of different-sized mesh in each net was randomly allocated.

Table 1
Total numbers of individuals of the 10 most abundant species sampled by multi-mesh gillnets in shallow and deep strata in Lake Macquarie and St Georges Basin

	No.	%
Lake Macquarie		
Shallow		
Herklotsichthys castelnaui	763	32.1
Girella tricuspidata	324	13.6
Gerres subfasciatus	319	13.4
Liza argentea	217	9.1
Mugil cephalus	132	5.6
Rhabdosargus sarba	91	3.8
Pomatomus saltatrix	75	3.2
Pelates sexlineatus	68	2.9
Acanthopagrus australis	65	2.7
Platycephalus fuscus	60	2.5
All other 22 spp.	260	11.0
Total	2374	100
Deep		
G. subfasciatus	289	35.0
P. sexlineatus	110	13.3
H. castelnaui	96	11.6
Sillago maculata	86	10.4
P. saltatrix	68	8.2
A. australis	58	7.0
L. argentea	24	2.9
Dinolestes lewini	21	2.5
P. fuscus	14	1.7
Pagrus auratus	14	1.7
All other 14 spp.	46	5.6
Total	826	100
St Georges Basin		
Shallow		
Sillago ciliatia	604	26.3
L. argentea	361	15.7
M. cephalus	306	13.3
Myxus elongatus	295	12.8
G. subfasciatus	224	9.8
G. tricuspidata	137	6.0
P. saltatrix	88	3.8
Pseudocaranx dentex	78	3.4
A. australis	69	3.0
P. fuscus	51	2.2
All other 14 spp.	84	3.7
Total	2297	100
Deep		
A. australis	218	25.8
S. ciliatia	142	16.8
P. saltatrix	132	15.6
G. subfasciatus	110	13.0
R. sarba	64	7.6
P. auratus	40	4.7
G. tricuspidata	38	4.5
P. dentex	26	3.1
S. maculata	18	2.1
L. argentea	14	1.7
All other 7 spp.	42	5.0
Total	844	100

% = percent contribution to total sample.

2.3. Analyses of data

Data were analysed using standard multivariate and univariate techniques. Multivariate analyses were done on Bray-Curtis dissimilarity matrices and non-metric multidimensional scaling (nMDS) was used to display multivariate patterns of assemblages. Analyses of similarity (ANOSIM; Clarke, 1993) and permutational multivariate analysis of variance (PERMANOVA; Anderson, 2005) tested hypotheses about differences in assemblages between lakes, depths and at the different spatial scales (i.e. zones and sites). Analyses were done using: (i) untransformed abundance data (to test hypotheses about the composition and abundance of species in the

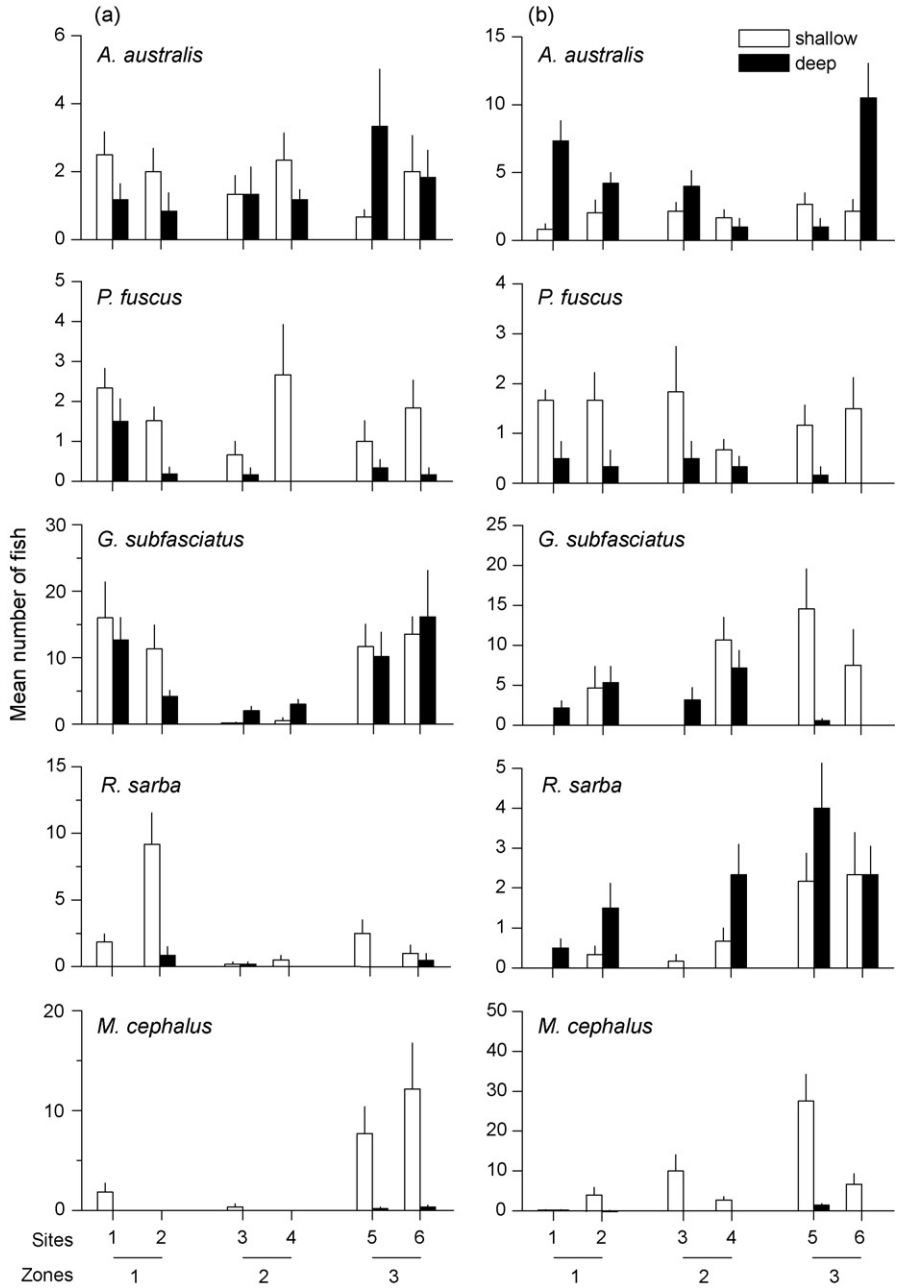

Fig. 2. Mean (+S.E.; $n = 6$) numbers of common species of fish sampled with multi-mesh gillnets in shallow and deep strata at six sites (nested in three zones) in (a) Lake Macquarie and (b) St Georges Basin.

assemblages); (ii) standardised data (to determine whether species were sampled in similar proportions in different samples); (iii) presence/absence data (to test hypotheses about the composition of assemblages and frequencies of occurrence of species).

Abundances of individual species were analysed separately for each lake and depth using nested ANOVA to obtain independent measures of components of variation for each spatial scale (Underwood, 1997). Analyses were restricted to species that were caught in more than 25% of samples at each depth in each lake. All spatial factors were considered random; analyses were done on untransformed data. Components of variation were calculated for each spatial scale (zones, sites and replicates) as in Underwood (1997). Negative components of variation were removed using pooling procedures (Fletcher and Underwood, 2002). Multivariate components of variation were similarly calculated for each of the different spatial scales from the PERMANOVA analyses.

Size-frequency distributions of abundant species (>50 individuals in both shallow and deep samples pooled across sites) were compared between depths in each lake using Kolmogorov–Smirnov (K–S) tests.

3. Results

3.1. General

Similar numbers of fish were sampled across depths and lakes, but more species were sampled at both depths in LM than in SGB

Table 2
Results of components of variance derived from ANOVA for common species sampled at each depth in Lake Macquarie and St Georges Basin

	Lake Macquarie				St Georges Basin			
	Deep		Shallow		Deep		Shallow	
	VC	ø	VC	ø	VC	ø	VC	ø
A. australis								
Zones	0.3	0.06	0	–	12.1	1.02	0	–
Sites	0.0	–	0	–	1.4	0.12	0	–
Res	4.5	–	3	–	11.8	–	3.2	–
G. subfasciatus								
Zones	19.5	0.26	50	0.87	4.2	0.35	4	0.07
Sites	5.7	0.08	0	–	2.4	0.2	20.8	0.35
Res	75	–	57.5	–	12	–	60	–
G. triscuspidata								
Zones	na	na	0	–	2.5	0.64	0	0
Sites	na	na	0	–	0	–	4	0.67
Res	na	na	118.2	–	3.9	–	6	–
L. argentea								
Zones	0	–	0	–	0.2	0.5	129. 9	0.88
Sites	0.8	0.36	26.7	0.72	0	–	20.8	0.14
Res	2.2	–	37.2	–	0.4	–	146.9	–
M. cephalus								
Zones	na	na	27	0.94	na	na	17.1	0.23
Sites	na	na	0	–	na	na	70.6	0.96
Res	na	na	28.7	–	na	na	73.4	–
P. fuscus								
Zones	0	–	0	–	na	na	0	–
Sites	0.2	0.5	0.1	0.04	na	na	0	–
Res	0.4	–	2.8	–	na	na	1.7	–
P. saltatrix								
Zones	1.4	0.26	0	–	1.4	0.12	0	–
Sites	1.4	0.26	3.3	1	1	0.09	2.4	0.36
Res	5.3		3.3	–	11.6	–	6.6	–
R. sarba								
Zones	0	–	2.5	0.33	0.7	0.25	1.2	0.71
Sites	0	–	8.1	1.08	1.1	0.39	0	–
Res	0.7	–	7.5	–	2.8	–	1.7	–
S. ciliata								
Zones	na	na	0	–	0	–	25	0.2
Sites	na	na	0.8	0.42	1.3	0.18	121.7	1.4
Res	na	na	1.9	–	7.2	–	86.6	–
S. maculata								
Zones	0	–	0	–	0.2	0.4	na	na
Sites	0.9	0.16	0	–	0	–	na	na
Res	5.7	–	2.1	–	0.5	–	na	na
Multivariate								
Zones	232.6	0.12	419	0.27	483.1	0.43	124.8	
Sites	477.2	0.26	1045.5	0.68	347.4	0.31	901.5	
Res	1863.3	–	1541.6	–	1117.6	–	1078.7	

VC = variance component; Ø = the ratio estimated magnitude of variance: estimated residual variance; na = no analysis.

(Table 1). More individuals and species were sampled in shallow than in deep water for both lakes. Samples were dominated by relatively few species; the 10 most abundant species accounted for 89–96% of all individuals sampled (Table 1).

3.2. Differences in assemblages between lakes and depths

The structure and composition of assemblages differed significantly between lakes and depths within each lake (ANOSIM, $P < 0.05$; see Fig. 1). Some species were caught in large numbers and proportions in only one lake (e.g. *Herklotsichthys castelnaui*, *Sillago maculata* and *Pelates sexlineatus* in LM; *Sillago ciliatia* and *Myxus elongatus* in SGB (Table 1), but most species found in only one lake were caught in small numbers (e.g. *Macquaria colonorum*, *Hyporhamphus regularis ardelio*).

Assemblages differed between depths primarily because of species being more abundant in shallow samples. In each lake, 10 species accounted for a very large percentage of the dissimilarity between depths (82% in LM; 91% in SGB from SIMPER analyses, Clarke, 1993). Eight of the species in the set of 10 from each lake were more abundant in shallow than deep samples.

3.3. Scales of spatial variation

Abundances of common species varied between depths and among zones and sites within each estuary (Fig. 2). Separate analyses of spatial components of variation were therefore done for individual species and the assemblages for each depth in each lake, separately. Of the 38 analyses, 16 had negative components of variation at the scale of zones, 13 at the scale of sites and 6 at more than 1

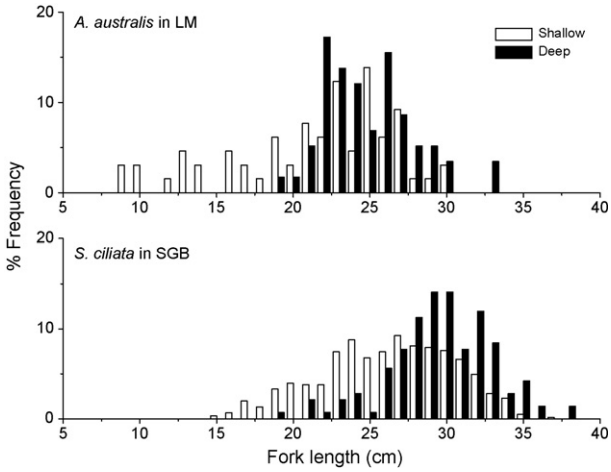

Fig. 3. Examples of fish sampled with multi-mesh gillnets that differed significantly in length between depths.

spatial scale (Table 2). The negative sources of variation were pooled (shown as zero variance in Table 2) and the other components were recalculated.

Most variation was at the smallest spatial scale of replicate nets (residual variance; Table 2) across species, depths and lakes for univariate and for multivariate data. Further, when components were scaled against the residual (to assess the magnitude of variation of each factor relative to the same reference; see Underwood, 1997), variation among zones was generally smaller than among sites. Not all species, however, displayed similar patterns of variation (e.g. *Gerres subfasciatus*; Table 2); nor were patterns among zones and sites consistent for all species across depths (e.g. *Acanthopagrus australis*) or lakes (e.g. *Liza argentea*).

There were significant differences (Permanova, $P < 0.05$) in the composition, abundance and proportions of species in assemblages (raw and standardised data) and the frequency of occurrence of species (presence/absence data) among sites for each depth and lake, but not between zones.

3.4. Size frequencies of fish

K–S tests detected significant differences in the size frequencies of *G. subfasciatus*, *A. australis* and *Pomatomus saltatrix* between depths in LM and of *S. ciliatia* in SGB (examples in Fig. 3). Greater proportions of smaller individuals were generally caught in shallow water (e.g. *G. subfasciatus* <12 cm, *A. australis* <20 cm and *S. ciliate* <25 cm FL). Greater proportions of larger individuals (*S. ciliatia* >27 cm and *P. saltatrix* >42 cm FL) were caught in the deep strata.

4. Discussion

The fish sampled here included several species (e.g. *G. subfasciatus*, *A. australis*, *Mugil cephalus* and *Girella tricuspidata*) that are important in the commercial estuarine fisheries of NSW (Gray, 2002; Gray and Kennelly, 2003). Our sampling with multi-mesh gillnets has provided additional data on (i) smaller sizes of these species; (ii) species not retained by commercial fishers (e.g. important baitfish species such as *H. castelnaui*) and (iii) species important in the recreational fishery (e.g. *P. saltatrix*).

The greater diversity of fish fauna in LM than in SGB accords with commercial beach-seine catches in these lakes (Gray and Kennelly, 2003). Similarly, the predominance of certain species in one lake, such as *S. maculata* in LM and *S. ciliatia* in SGB, is consistent with the beach-seine fisheries (Gray and Kennelly, 2003). Our sampling was not designed to ascertain the mechanisms and processes that caused the observed differences, but our results reinforce previous hypotheses that there are intrinsic dissimilarities in the fish fauna of these lakes. Similar patterns have been observed for benthic invertebrate fauna in coastal lakes (Dye and Barros, 2005). Thus, no particular lake could be classified or used as a 'reference' or 'control' for comparative purposes (see also Sheaves, 2006).

An important result was that most variability occurred at the smallest spatial scale investigated (among replicate nets), a pattern that was consistent across different species, depths and lakes. Another important result was that variation was generally larger among sites than among zones, although this pattern was not the same for all species, depths and lakes. Small-scale patchiness has also been observed for reef fishes in different habitats (Anderson and Millar, 2004) and is a general property of benthic assemblages in coastal habitats (review Fraschetti et al., 2005). Our data support the hypothesis that small-scale variability is common for a wide range of taxa in estuaries (Livingston, 1987). A similar understanding of scales of temporal variation (e.g. days, weeks, months and seasons) is required in designing more reliable long-term surveys (Morrisey et al., 1992).

Overall, our data show that including the scale of zones in future sampling is unnecessary; more effort should be put into sampling replicate nets and sites in future surveys of coastal lakes. Unless replication is sufficient at the relevant scales, future surveys are unlikely to be reliable for assessing patterns of abundance and detecting environmental impacts on assemblages of commercially important fish (Underwood and Chapman, 2003). In addition to identifying the scales at which the populations show most variation, the components of variation can also be used to do cost-benefit analyses, which will determine the most cost-efficient and optimal levels of spatial replication for future surveys (e.g. Kennelly et al., 1993; Underwood, 1997). This is the next step in our strategy to develop fishery-independent sampling tools (Rotherham et al., 2007). Pilot experiments are essential for designing cost-efficient and robust sampling strategies; here we have provided an example of the type of pilot studies that need to be done to assist in designing future research surveys.

Acknowledgements

This work forms part of a collaborative study between the Wild Fisheries Program (NSW Department of Primary Industries), the Centre for Research on Ecological Impacts of Coastal Cities (University of Sydney) and the Australian Fisheries Research and Development Corporation (Project 2002/059). Sampling was done under NSW Agriculture Animal Care and Ethics approval 02/15. We thank Paul Lokys for assistance with sampling and Dr. Steve Kennelly for comments on the manuscript.

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